

PROJECT AND TEAM INFORMATION

Project Title :

NEXT MOVE: Warehouse Route Optimizer

Student / Team Information

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PROPOSAL DESCRIPTION (10 pts)

Motivation

In a warehouse, products are stored across different zones, aisles, racks, shelves and bins. When a customer places an order containing several products, a worker may need to visit multiple locations. Without an organized picking strategy, the worker can travel unnecessary distance and spend more time completing the order. A second problem is inventory mismatch: a product may appear available in the application but be unavailable at its actual warehouse location. This can lead to unwanted substitutions, order changes and lower customer satisfaction. A mismatch can also affect free-delivery eligibility when the unavailable item causes the cart value to fall below the required threshold. NEXT MOVE addresses these issues through inventory validation, alternative product suggestions, order management, warehouse mapping and dynamic pick-path optimization. The system will guide workers to exact product locations and update the route when warehouse conditions change.

State of the Art / Current solution

Current warehouse management systems can manage inventory, orders, product locations and picking operations. Graph-based routing and shortest-path algorithms are also used to reduce picker travel. Mobile or handheld picking systems can provide workers with product lists, locations and operational instructions. However, industrial systems may depend on large-scale infrastructure, databases, scanners, sensors or proprietary platforms. Our project focuses on a compact, explainable academic prototype centred on Data Structures and Algorithms using C and C++, with a worker-oriented interface.

Project Goals and Milestones

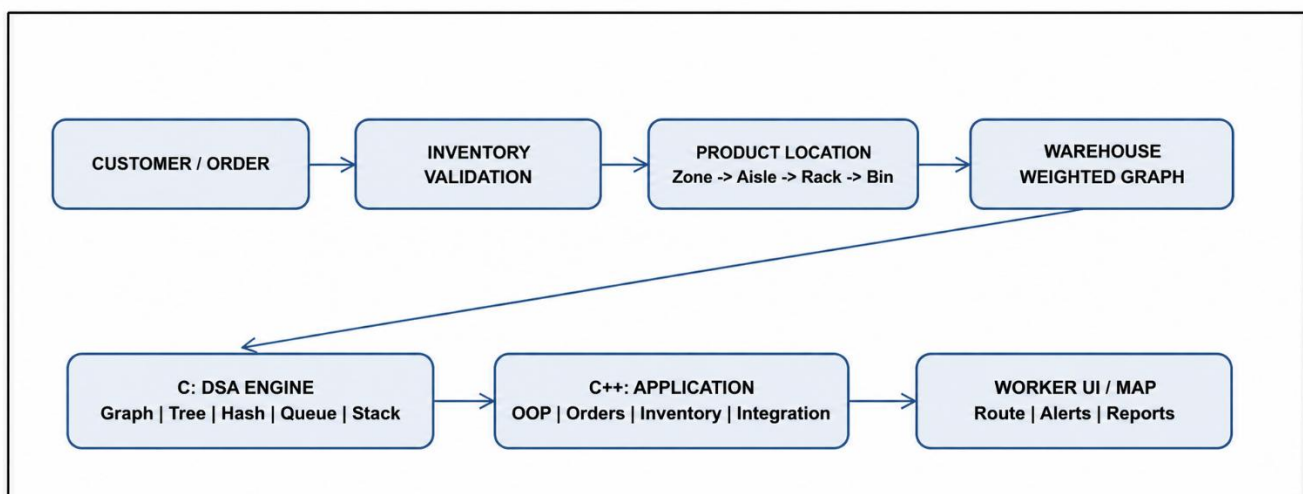
The goal is to develop a working prototype connecting customer order handling, inventory validation and warehouse picking optimization. Initial milestones are:

- Define the warehouse and product data model.
- Implement core Data Structures in C.
- Implement product, inventory and order modules.
- Build the warehouse graph and implement BFS, DFS and Dijkstra.
- Develop picking-sequence and dynamic re-routing logic.
- Integrate the C modules with the C++ object-oriented application.
- Develop the worker interface and map visualization with permitted AI-assisted UI tools.
- Test the system and compare normal versus optimized picking distance and estimated time.

Project Approach

The solution will use two connected layers. The C layer will implement the core Data Structures and algorithms: arrays, linked lists, stacks, queues, priority queues, trees, hashing and graphs, together with searching, sorting, BFS, DFS, Dijkstra and path reconstruction. The C++ layer will model Product, Order, Warehouse, Zone, Aisle, Rack, Shelf, Bin, Inventory and Worker using classes and will coordinate application flow. The warehouse will be represented as a configurable weighted graph. Product locations will follow Zone -> Aisle/Row -> Rack -> Shelf -> Bin. Hashing will provide fast product lookup; a tree will organize categories; queues will manage normal orders; priority queues will handle urgent orders; and the graph will support navigation. For multiple products, an efficient picking sequence will be generated and Dijkstra will calculate shortest weighted paths between consecutive stops. If a path becomes blocked, the graph will be updated and the remaining route recalculated. The final interface will show orders, inventory, map, route, alerts and reports.

SYSTEM ARCHITECTURE (HIGH LEVEL DIAGRAM)



Architecture note: C contains the core Data Structures and algorithms. C++ models the warehouse entities and application flow. The worker interface displays orders, inventory, map routes, alerts and reports. When an aisle/path is blocked, the graph is updated and the remaining route is recalculated.

PROJECT OUTCOME / DELIVERABLES

The deliverable will be a working prototype of NEXT MOVE demonstrating an end-to-end warehouse picking workflow. It will include product and inventory management, availability validation, alternative-product suggestions, order and priority management, exact warehouse location handling, a configurable warehouse graph, BFS/DFS traversal, Dijkstra-based shortest-path calculation, efficient picking-sequence generation, dynamic re-routing for blocked paths, and a worker-oriented map/interface. The prototype will also provide normal-versus-optimized distance/time comparison and basic operational reports. The primary academic outcome is a demonstrable C/C++ implementation showing practical use of Data Structures and Algorithms in a real-world warehouse problem.

ASSUMPTIONS

The warehouse layout and product-location information are provided in structured form. A product may be identified by Zone, Aisle/Row, Rack, Shelf and Bin. For the prototype, worker position can be simulated or updated through defined checkpoints; automatic indoor GPS-level positioning is not assumed. Alternative recommendations are rule-based rather than ML-based. Route optimization uses non-negative path costs and configurable warehouse conditions.

REFERENCES

- T. H. Cormen, C. E. Leiserson, R. L. Rivest and C. Stein, *Introduction to Algorithms*, MIT Press.
- E. Horowitz, S. Sahni and S. Anderson-Freed, *Fundamentals of Data Structures*.
- B. Stroustrup, *The C++ Programming Language*, Addison-Wesley.
- E. W. Dijkstra, "A Note on Two Problems in Connexion with Graphs", *Numerische Mathematik*, 1959.
- Course and laboratory resources for Data Structures and Programming in C++.